compute descriptive statistics for a real dataset; identify skewed distributions; apply zscore normalization; compute pearson and spearman correlation

In [17]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load Titanic dataset
df = sns.load_dataset('tips')
#df = pd.read_json("titanic.json")

# Descriptive statistics
stats = df.describe()

print(stats)
```

```
       total_bill         tip        size
count  244.000000  244.000000  244.000000
mean    19.785943    2.998279    2.569672
std      8.902412    1.383638    0.951100
min      3.070000    1.000000    1.000000
25%     13.347500    2.000000    2.000000
50%     17.795000    2.900000    2.000000
75%     24.127500    3.562500    3.000000
max     50.810000   10.000000    6.000000
```

In [18]:

```
# Select numeric columns
num_cols = df.select_dtypes(include='number')

# Calculate skewness
skewness = num_cols.skew()
print("Skewness of Numeric Columns:")
print(skewness)
```

```
Skewness of Numeric Columns:
total_bill    1.133213
tip           1.465451
size          1.447882
dtype: float64
```

In [22]:

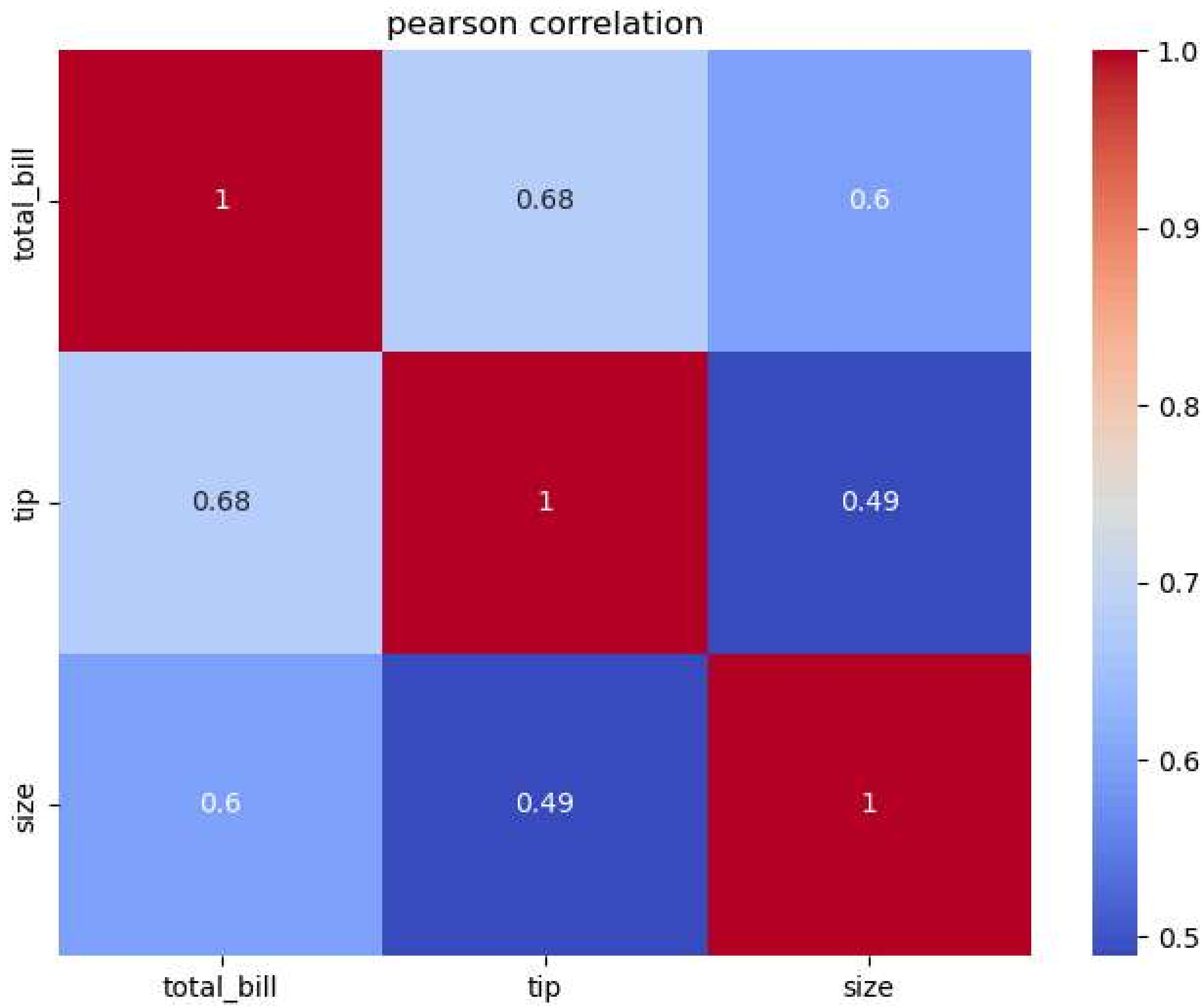
```
# Apply Z-score normalization
zscore_df = num_cols.copy()
for col in zscore_df.columns:
    zscore_df[col]=(
        zscore_df[col]-zscore_df[col].mean()
    )/zscore_df[col].std()
print(zscore_df.head())
```

```
   total_bill       tip      size
0   -0.314066 -1.436993 -0.598961
1   -1.061054 -0.967217  0.452453
2    0.137497  0.362610  0.452453
3    0.437416  0.225291 -0.598961
4    0.539635  0.442111  1.503867
```

In [26]:

```
# Pearson Correlation
pearson_corr = num_cols.corr(method='pearson')
print(pearson_corr)
#Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='coolwarm'
)
plt.title("pearson correlation")
plt.show()
```

```
            total_bill       tip      size
total_bill    1.000000  0.675734  0.598315
tip           0.675734  1.000000  0.489299
size          0.598315  0.489299  1.000000
```

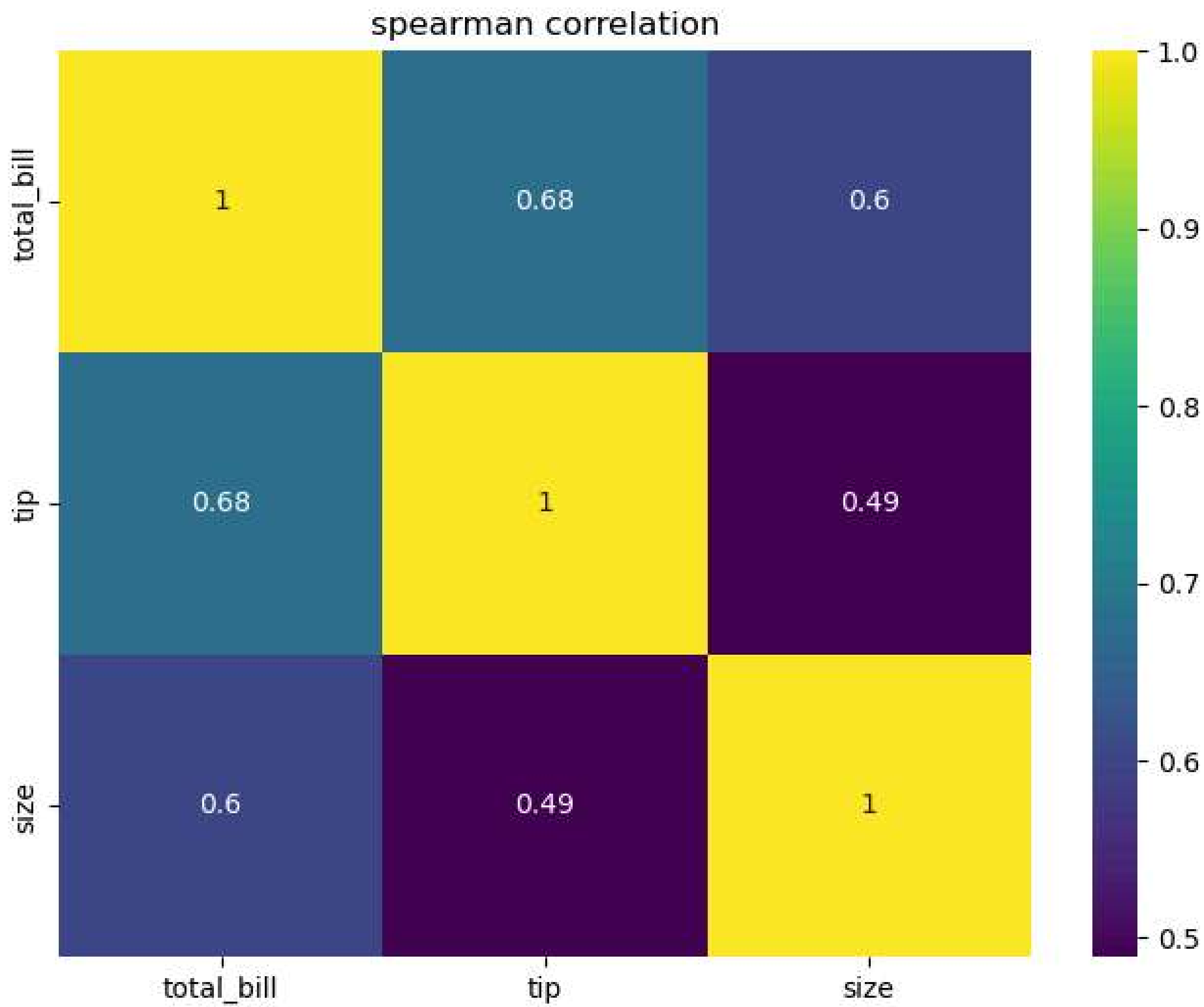


In [29]:

```
# Spearman Correlation
spearman_corr = num_cols.corr(method='spearman')
print(spearman_corr)
#Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(
    pearson_corr,
    annot=True,
    cmap='viridis'
)
```

```python
plt.title("spearman correlation")
plt.show()
```

```
            total_bill       tip      size
total_bill    1.000000  0.678968  0.604791
tip           0.678968  1.000000  0.468268
size          0.604791  0.468268  1.000000
```



In [38]:
```python
#compare pearson and spearman
comparison = pd.DataFrame({
    "pearson": pearson_corr["total_bill"],
```

```python
    "spearsman": spearman_corr["total_bill"]
})
print(comparison)
```

```
            pearson  spearsman
total_bill  1.000000   1.000000
tip         0.675734   0.678968
size        0.598315   0.604791
```

In [ ]:

```python

```